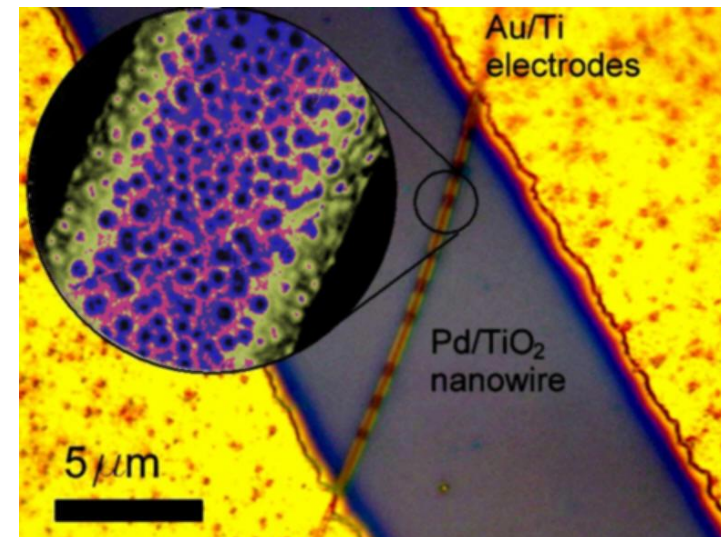


Topic 4: One-Dimensional Nanostructures

Questions we will answer...

- How are nanowires synthesized?
- What are the fundamental processes of vapor-phase growth?
- What are evaporation-condensation and dissolution-condensation processes?
- What are the VLS & SLS process?
- What are alternative methods of nanowire growth?



Source: Kolmakov, SIUC

One-Dimensional Nanostructures are...

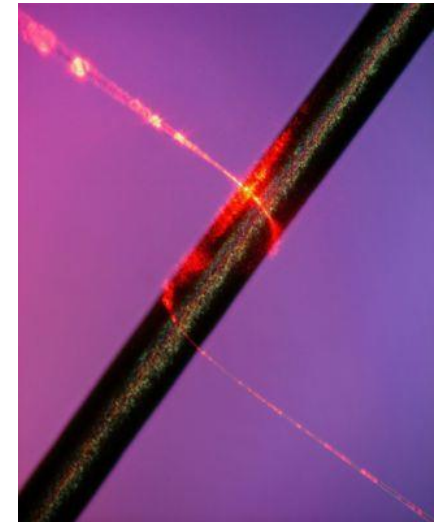
Whiskers, fibers, fibrils, nanowires, nanorods.

Nanotubes, nanotubules, nanocables also considered 1-D nanostructures.

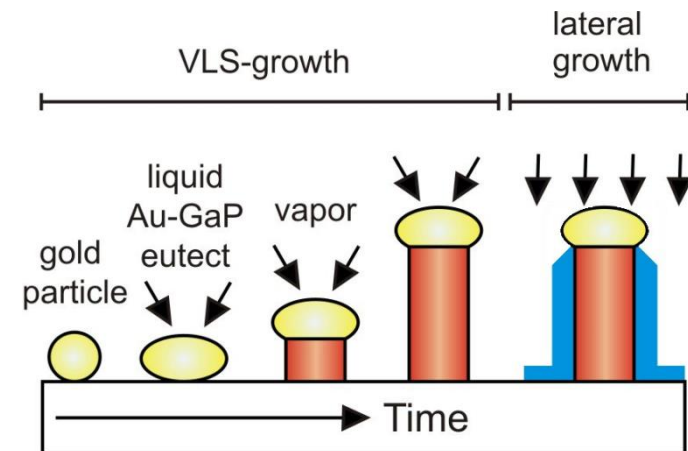
Whiskers and fibers come to us from before the “nano” branding age; these referred to 1-D structures with diameters ranging from nanometers to several hundred microns.

Nanowires and nanorods are terms used to describe 1-D nanostructures with diameters smaller than several hundred nanometers.

Much of the science behind the growth of nanowires and nanorods comes to us from the early work on whiskers and fibers.



Silica nanowire



Source: Philips Research

Techniques to Synthesize Nanowires

Spontaneous growth

- Evaporation (or dissolution)-condensation
- Vapor (or solution)-liquid-solid (VLS or SLS) growth
- Stress-induced recrystallization

Template-based synthesis

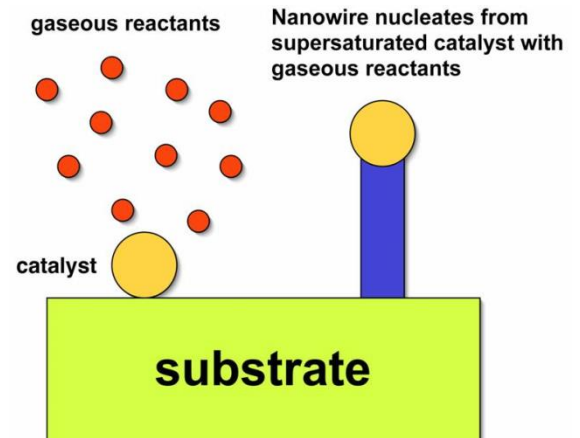
- Electroplating and electrophoretic deposition
- Colloid dispersion, melt, or solution filling
- Conversion with chemical reaction

Electrospinning

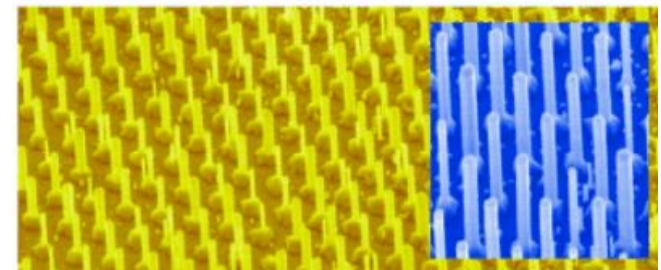
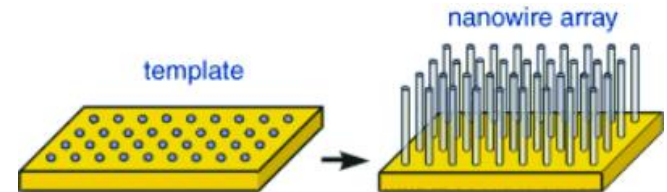
Lithography

Top-down

Bottom-up



Source: TASC, Italy



Zacharias, Germany

Spontaneous Growth

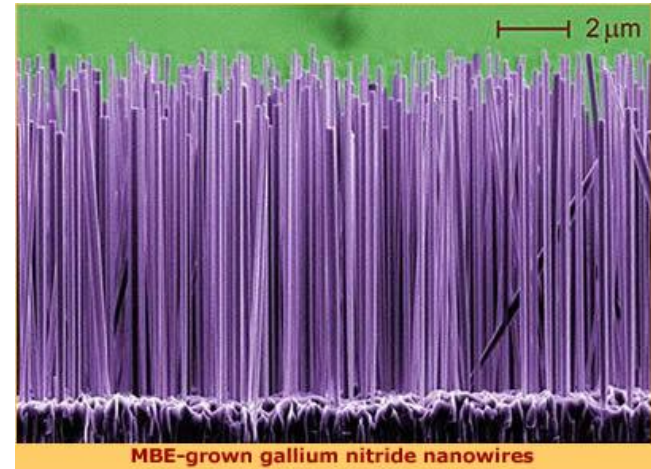
Thermodynamically driven process $\Leftrightarrow$
reduction of the Gibbs free energy *via*...

- Phase transformation (heterogeneous nucleation & growth)
- Chemical reaction
- Stress release

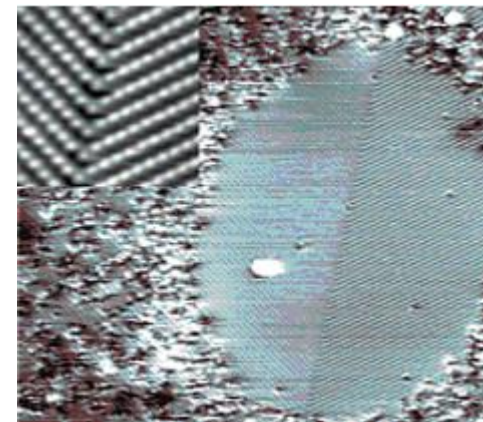
Nanowire/nanorod formation requires anisotropic growth, i.e., crystal growth faster along a certain growth direction than another; or else $\rightarrow$ 0-D nanostructures!

Uniformly sized nanowires/nanorods form if growth is stunted along all directions except the growth direction

Growth conditions, defects and surface morphology play an important role in determining end-product of spontaneous growth.



NIST



Mikkelsen; Lund University;
Sweden; nanowire consisting of
two nanocrystallites viewed by
STM after cleavage. Topic 4 - 4

Evaporation (dissolution)-condensation growth

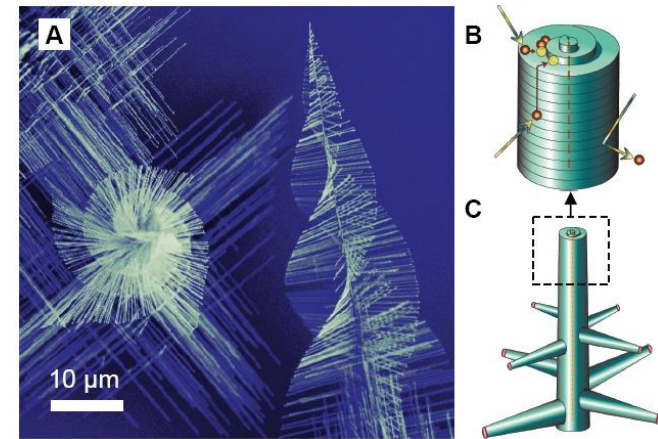
Fundamentals

Evaporation-condensation process, AKA vapor-solid (VS) process

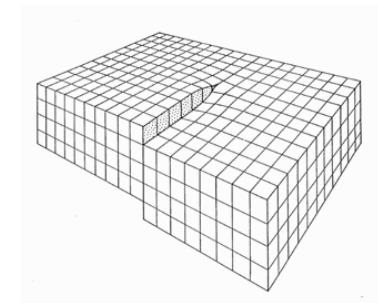
Driving force is thermodynamic (reduction of the Gibbs free energy) from recrystallization or decrease in supersaturation → nanowires and nanorods are typically single-crystalline, low in defects and imperfections

Main mechanism: anisotropic growth; How?

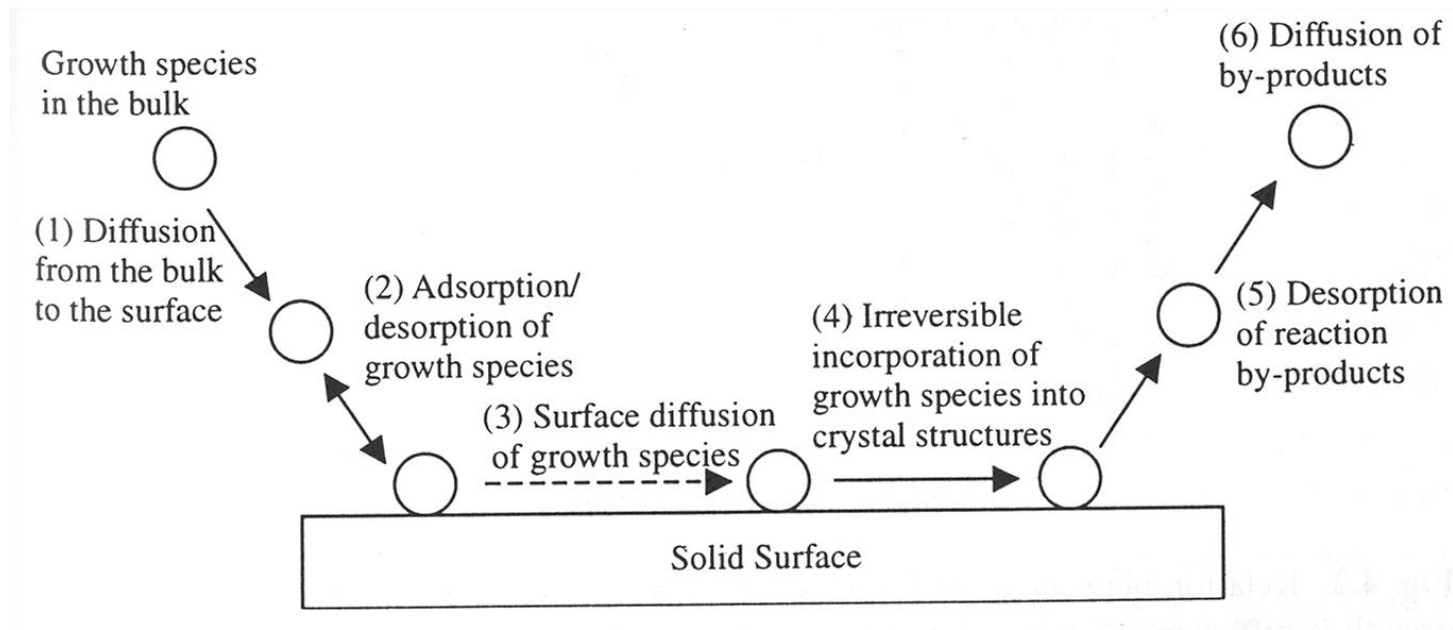
- Different facets in a crystal have different growth rate (e.g., in Si: $\{111\}$ facets grow slower than $\{110\}$)
- Presence of imperfections in specific growth directions (e.g., screw dislocations) can cause anisotropic growth
- Preferential accumulation of impurities and poisoning on specific facets (e.g., oxidation)
- Preferential ligand coverage of specific facets in solution



Jin group; U of Wisconsin-Madison



Fundamentals of Crystal Growth



1. Growth species diffuse from vapor or liquid to the growth surface (not rate-limiting)
2. Growth species adsorb/desorb [**can be rate-limiting**; → determines growth rate ($\text{nm}\cdot\text{s}^{-1}$)]
3. Surface diffusion of adsorbed growth species (desorption is still possible, so is irreversible incorporation)
4. Surface growth by irreversible incorporation into the crystal structure [**can be rate-limiting** process → determines growth rate ($\text{nm}\cdot\text{s}^{-1}$)]
5. If reactive process, then chemical by-products generated during growth are expected to desorb from the growth surface, so that growth species can adsorb onto the surface and the process can continue.

Adsorption/Diffusion-Mediated Crystal Growth (step 2)

If the rate-limiting step is adsorption-desorption, then growth rate is determined by condensation rate, J (atoms- $\text{cm}^{-2}\text{-s}^{-1}$)

– J depends on...

- # of growth species, which is proportional to vapor pressure or concentration, P , of growth species in the vapor

$$J = \frac{\alpha \sigma P_0}{\sqrt{2\pi m k T}}$$

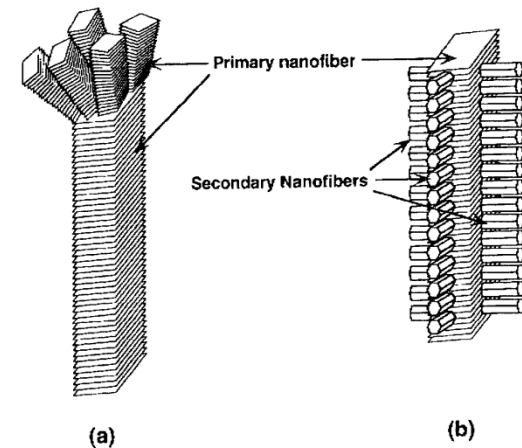
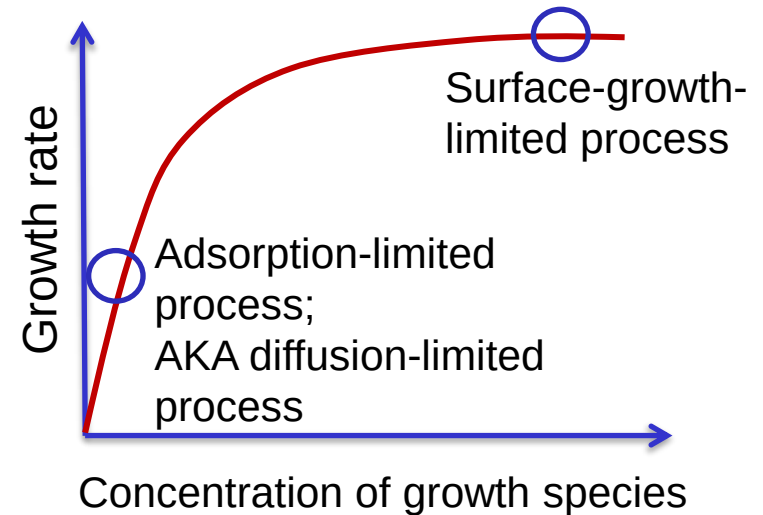
$$\sigma = \frac{P - P_0}{P_0}$$

- α : accommodation coefficient (fraction of species accommodated by the surface; surface-specific); σ : supersaturation; P_0 : equilibrium vapor pressure at temperature T , and m : atomic weight of growth species
- Accommodation is a thermalization/attachment process; high $\alpha \rightarrow$ high growth rate; different α on different facets $\rightarrow$ anisotropic growth

Growth Regimes

Low P_0 (C_0) $\rightarrow$ supply and adsorption are more likely to be the rate-limiting steps
Growth rate increases linearly with P_0 (C_0) until there is a change from adsorption-limited to surface-growth-limited regimes
In surface-growth-limited (or surface-limited) regime, growth rate is independent of P_0 (C_0)

Surface-limited regime is far from equilibrium and results in formation of many defects and impurities, and even secondary nucleation or homogeneous nucleation (must be avoided usually, but can be harnessed!)



Kinetics of Surface Processes

Impinging growth species have a finite residence (or diffusion) time (τ_s) on the surface – during which they are *adatoms* – prior to escaping back to the vapor

$$\tau_s = \frac{1}{\nu} e^{\frac{E_{des}}{kT}}$$

- ν is the vibrational frequency of the adatom (10^{12} Hz), and E_{des} is the adsorption energy barrier

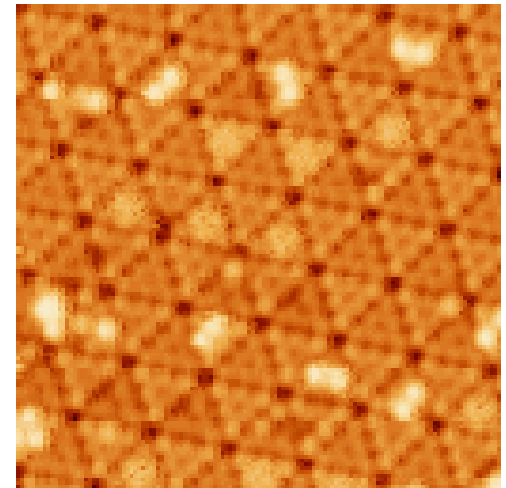
While on the surface, an adatom can diffuse along the surface with a diffusion coefficient D_s :

$$D_s = \frac{1}{2} a_0 \nu e^{-\frac{E_s}{kT}}$$

- E_s is the activation barrier to diffusion and a_0 is the size of the adatom

Mean diffusion distance, X , of the adatom from the site of incidence is given by

$$X = \sqrt{2D_s\tau_s}$$



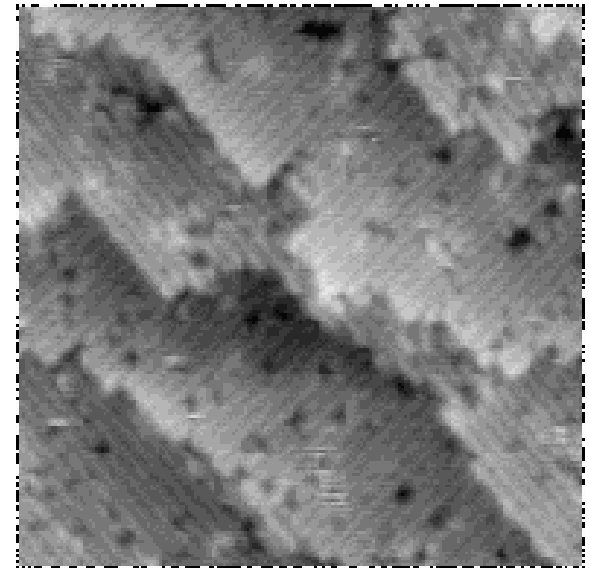
Pb on Si(111)-7x7 surface
Source: Gómez-Rodríguez

Kinetics of Surface Processes

If $X \gg$ distance between two growth sites (e.g., step edge, kink)
→ all adatoms can be accommodated (incorporated)

If $X \ll$ distance between growth sites then all adatoms can escape back to the vapor

Accommodation depends on E_s , E_{des} and density of growth sites
→ it is facet-specific, depends on temperature (surface morphology), and is material-specific



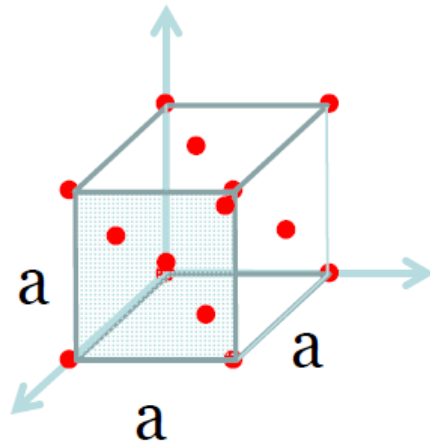
STM of Si(001); James Owen

Surface Growth-Limited Process (step 4)

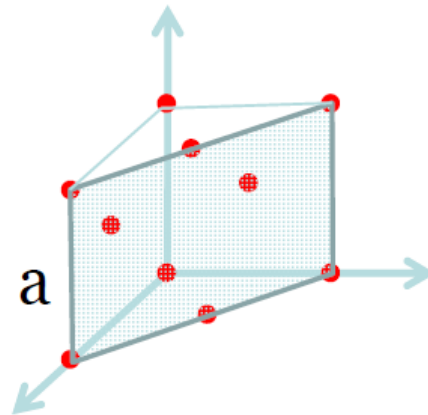
Rate limiting step is incorporation of adatoms into the surface

Reminder: different facets have different planar atomic density & different # of unsatisfied bonds (broken or dangling)
different surface energy, γ (e.g., FCC)

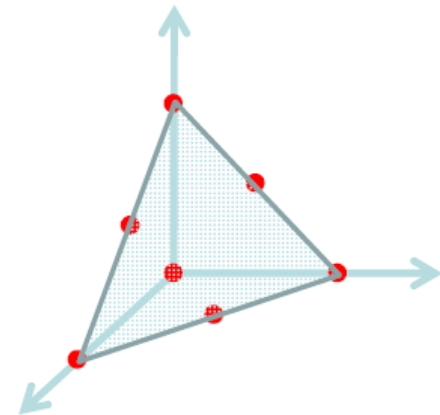
$$\rightarrow \gamma = \frac{N_b \epsilon \rho_a}{2}$$



$$\gamma_{(100)} = 4 \frac{\epsilon}{a^2}$$



$$\gamma_{(110)} = \frac{5}{\sqrt{2}} \frac{\epsilon}{a^2}$$



$$\gamma_{(111)} = 2\sqrt{3} \frac{\epsilon}{a^2}$$

Differences in γ or number of unsatisfied bonds, leads to different accommodation, growth mechanisms and growth rates

Growth on a Flat Surface (KSV theory)

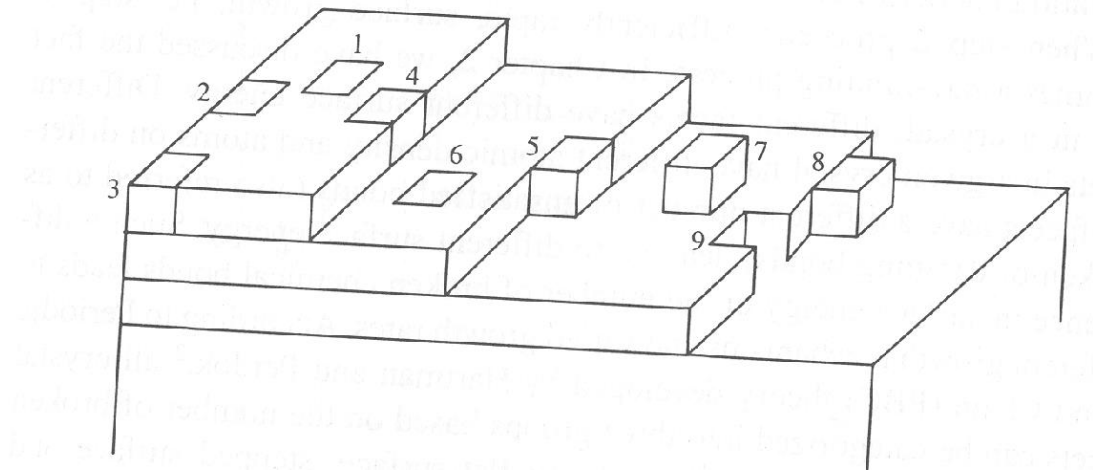
Classic step-growth theory developed by Kossel, Stranski and Volmer (KSV theory)

Premise:

- Crystal surfaces are not smooth, flat or continuous on the atomic scale (the scale that matters the most!)
- Discontinuities are responsible for crystal growth (they are incorporation sites or nucleation sites)

Illustration:

- $\{100\}$ surface of a simple cubic crystal of an elemental metal; each atom in the cube forms six chemical bonds (coordination number)



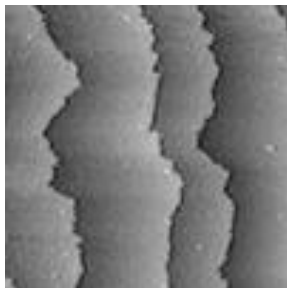
Growth on a Flat Surface (KSV theory)

Adatom (typically with one bond) diffuses randomly until it reaches an energetically favorable site → irreversible incorporation leading to growth

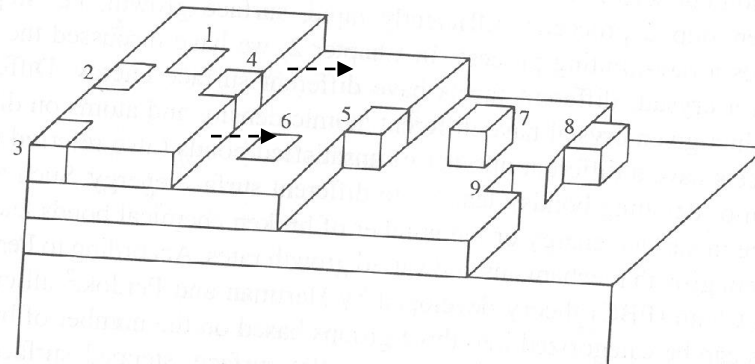
Different sites have different energy:

- On terrace → one bond (adatom) **(7)**
- At step edge (ledge site) → two bonds **(8)**
- At ledge-kink site → three bonds **(5)**
- At kink site → four bonds **(4)**
- In terrace → five bonds **(1 & 6)**
- When step edge buries terrace → six bonds

Growth proceeds by advancement of steps, AKA step flow



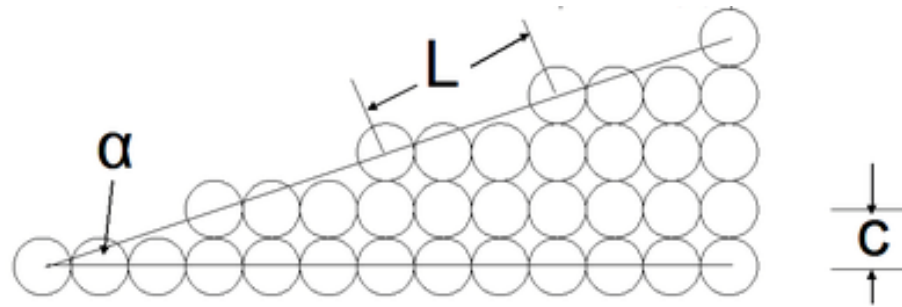
Si (001)



Growth on a Flat Surface (KSV theory)

For a given crystal facet and conditions, growth rate will depend on step (ledge) density

Miscut or misoriented surfaces favor incorporation of adatoms
→ faster growth rate



Limitation: **growth is limited by regeneration of growth sites**

Screw dislocation (BFC theory)

Burton, Cabrera, Frank (BCF theory): screw dislocation serves as continuous source of growth sites

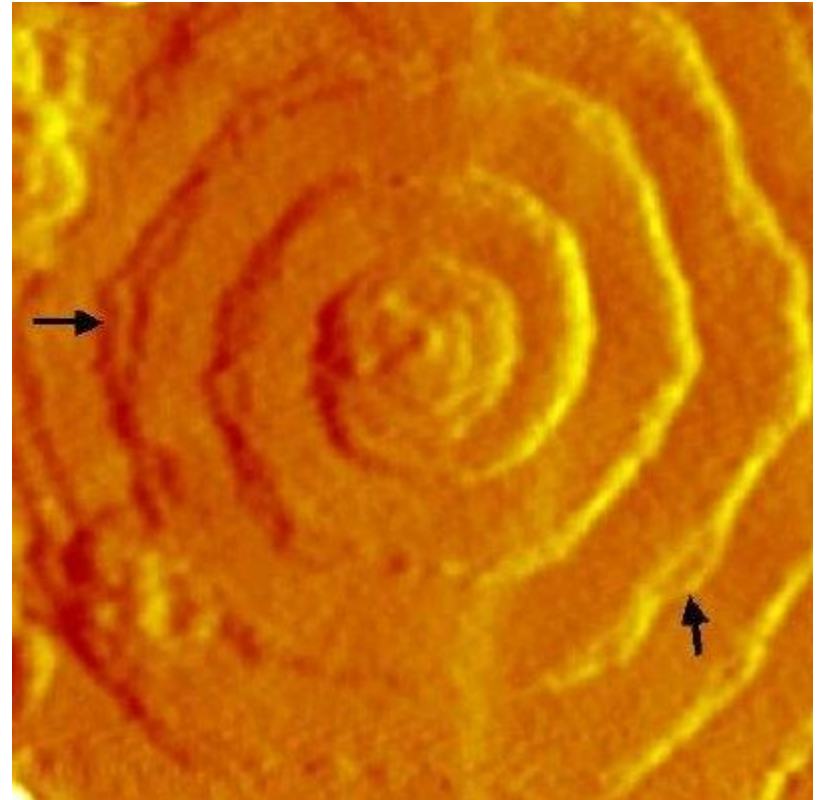
➔ stepped growth can continue

Screw dislocation ensures continuing advancement of growth surface

Different facets accommodate screw dislocations differently

Screw dislocation on a given facet

➔ anisotropic growth (formation of nanowires or nanorods)



Graphite screw dislocation; Rakovan

Periodic bond chain (PCB) theory

Developed by Hartman and Perdok

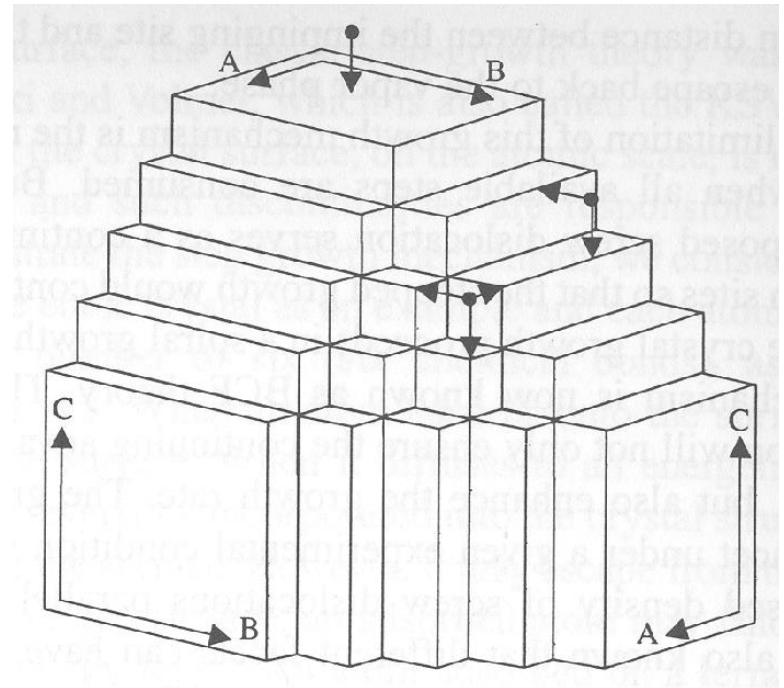
Premise: all crystal facets can be categorized into three groups:

- Flat surfaces
- Stepped surfaces
- Kinked surfaces

Difference between facets: number of broken periodic chains (# of broken bond per atom on a given facet)

Illustration:

- Assume a simple cubic crystal
- (100) → flat (F); (110) → stepped (S); (111) → kinked (K)
- (S) and (K) facets have faster growth rate in a simple cubic system



Anisotropic Growth

Generally:

- S and K have adsorption-limited growth, since accommodation is complete
- Accommodation in F ranges from 0 to 1, depending on availability of kink and ledge sites

Note: Facets with fast growth rate (high surface energy) tend to disappear, just as in a crystal in thermodynamic equilibrium (Wulff plot)

Consequence: formation of nanostructures with high aspect ratio is limited to materials with special crystal structures

Anisotropic growth is achieved at low supersaturation, such that P or C is higher than P_0 or C_0 (saturation pressure and concentration, respectively) for the growth surface and lower than P_0 or C_0 for non-growth surfaces

Also, anisotropic growth may require other mechanisms, such as defect-induced growth and/or impurity-inhibited growth

Evaporation-Condensation

Evaporation of growth species in vacuum, followed by condensation of supersaturated vapor on a cold surface

Mercury whiskers were the first to be grown (1-2 mm length with 200 nm diameter; yr: 1955)

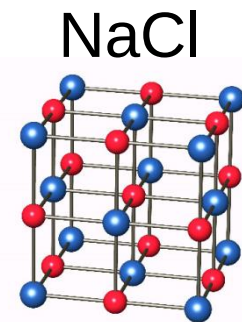
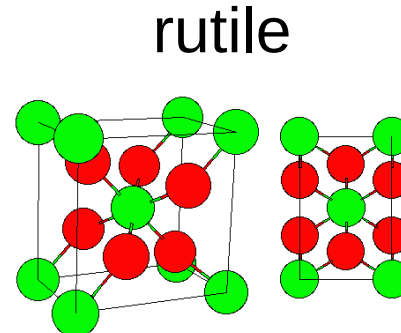
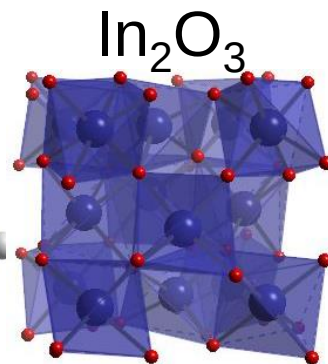
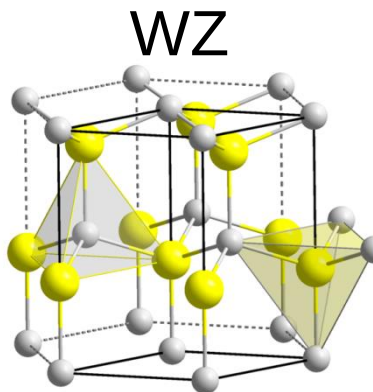
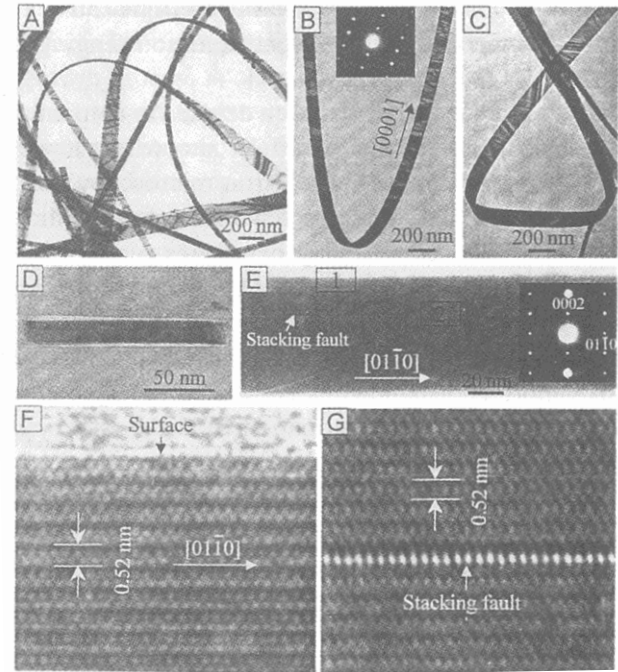
Origin of anisotropic growth:

- Axial screw dislocation believed to be at the origin of nanowire growth (not confirmed)
- Other possibility: micro-twins and stacking faults
- Still a topic of debate...

Evaporation-Condensation

Wang formed single-crystal nanobelts of semiconducting oxides:

- ZnO (wurtzite hexagonal)
(see image)
- In_2O_3 (C-rare-earth)
- SnO_2 (rutile)
- CdO (NaCl cubic)

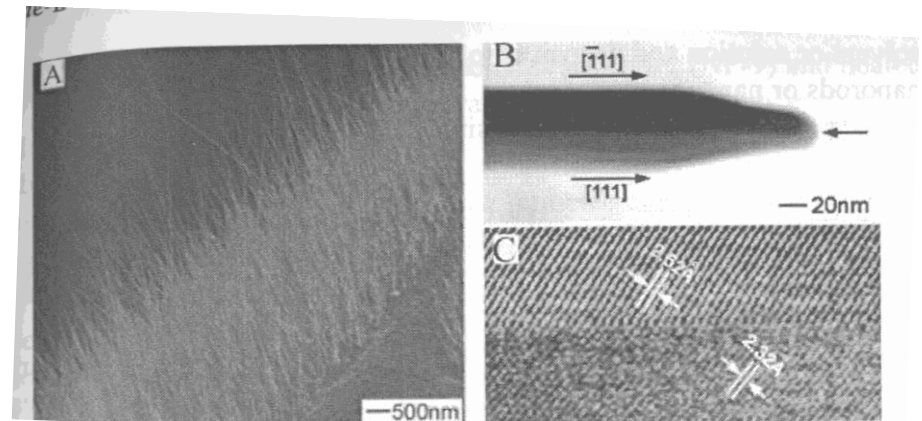
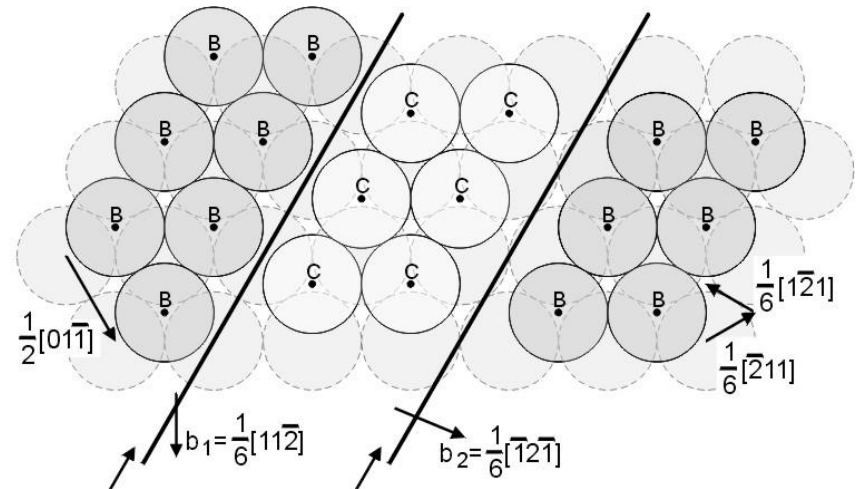


Evaporation-Condensation

Recipe: evaporate metal oxides at high T at 300 torr in alumina furnace and condense on an alumina substrate kept at slightly lower T

Note: no screw dislocations or impurities detected; all nanobelts have different structures; only stacking fault observed!

Role of stacking fault is up for debate.



CuO nanowires (bicrystals)

Recipe: heat Cu wire in air at 500°C for 4 hrs

Dissolution-Condensation

Differs from E-C only in growth media

Growth species first dissolve into solvent or solution, diffuse through liquid and deposit onto the surface

Growth via homogeneous or heterogeneous nucleation

Examples:

- Spherical colloids of amorphous Se (300 nm) prepared through reduction of selenious acid with excess hydrazine at 100°C; solution cooled to RT → Se nanocrystals (trigonal) precipitated; solution is aged → colloidal particles dissolve and nanocrystals grow anisotropically along the [001] direction
- Ag growth on Pt nanoparticles initiated by reduction of AgNO₃ with ethylene glycol; anisotropic growth induced by surfactant polyvinyl pyrrolidone (PVP; preferential attachment to some surfaces and poisoning)

Vapor (or solution)-Liquid-Solid (VLS or SLS) Growth

A secondary material (impurity or catalyst) used to direct/confine crystal growth

Role of catalyst:

- liquid drop traps and/or alloys with growth species
- Supersaturation of growth species → precipitation at growth surface, leading to 1D growth

First demonstration (45 yrs ago):

- Wagner fabricated Si nanowires/whiskers by VLS

Wagner's observations:

- No screw dislocations/imperfections in the growth direction
- In Si, $\langle 111 \rangle$ is the slowest direction as compared with low index surfaces, e.g., $\langle 110 \rangle$
- Impurities are always required
- A liquid-like globule is found at the tip of the nanowire

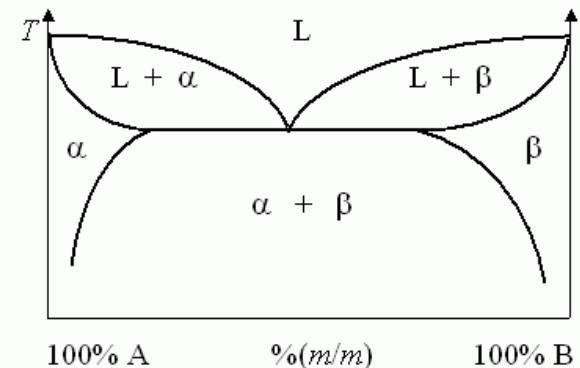
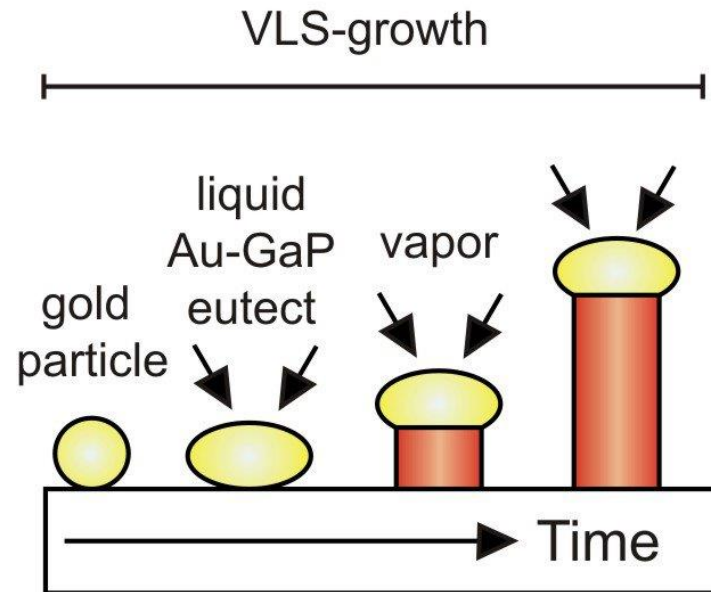
VLS Growth

Evaporated growth species
diffuses into a liquid droplet

The surface of the liquid is the
preferred site for deposition

Saturated growth species diffuse
to the edges and precipitate at
the solid-liquid interface
(nucleation and growth)

Continued growth leads to
nanowire formation



Principles of VLS Growth

Catalyst or impurity must form a liquid solution with growth species at deposition temperature

Distribution coefficient of catalyst < 1 at deposition temperature

Low equilibrium vapor pressure of catalyst is preferred, as loss of material could lead to reducing diameter of nanowire; catalyst can be supplied during growth

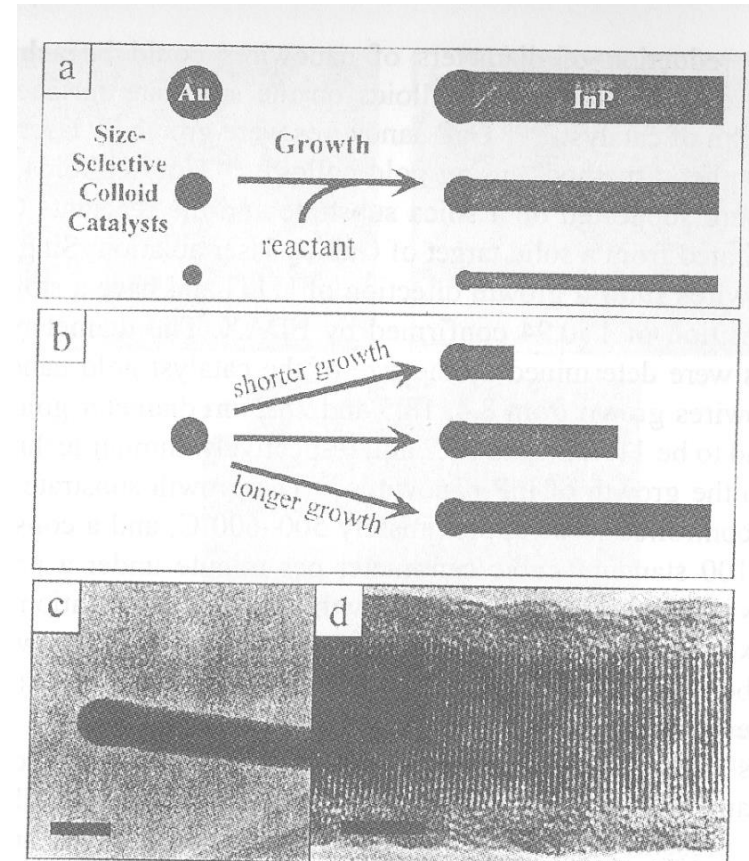
Catalyst must be chemically inert with regards to growth species and by-products

Interfacial energy is critical; wetting characteristics influence diameter. For a given volume, small wetting angle $\rightarrow$ large growth area and large diameter. Smaller volume $\rightarrow$ smaller diameter

Smaller volume $\rightarrow$ smaller diameter

For compound nanowire growth, one of the constituents can serve as catalyst

Unidirectional growth requires that solid-liquid interface have a well-defined crystallography (use single crystal substrates)



SLS Growth

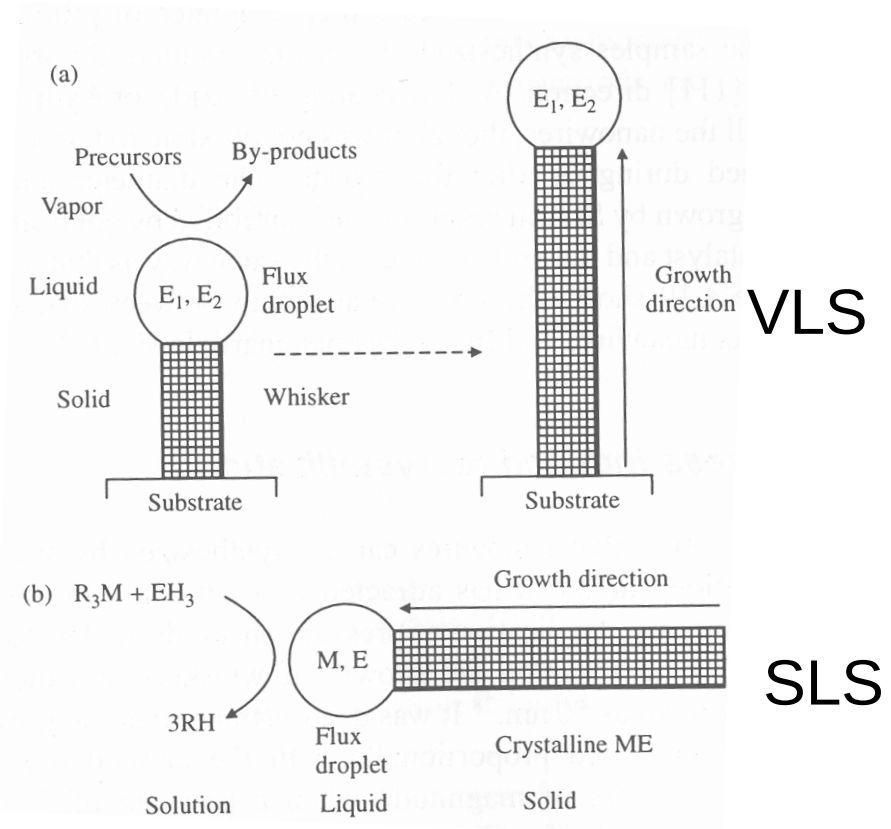
SLS is a solution-based method at lower temperature than VLS

Very similar to VLS

Compound nanowires, where one of the elements is the liquid catalyst

Example:

- InP nanowires
- In metal melts at 157°C and forms liquid drops
- $\text{In}(t\text{-Bu})_3 + \text{PH}_3 \rightarrow \text{InP} + 3(t\text{-Bu})\text{H}$
- In and P dissolve into In drop and precipitate to form InP nanowires



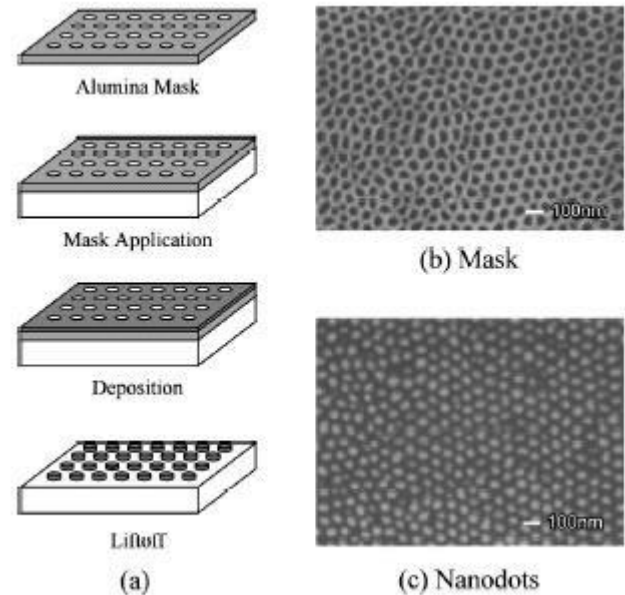
Template-Based Synthesis

General method applicable to most materials (polymers, metals, oxides, semiconductors) to form nanorods, nanowires, and nanotubules

Concept: templates are membranes with nanosized channels

Common membranes:

- Anodized alumina membranes
- Radiation track-etch polymer membranes
- Nanochannel array glass
- Mesoporous materials



Anodized Alumina

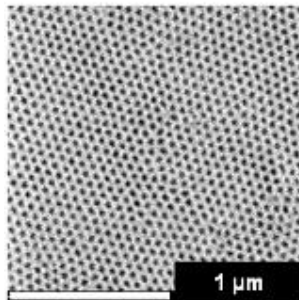
Anodic oxidation in sulfuric, oxalic, or phosphoric acids.

Yields hexagonally distributed pores

Pore size ~ 10 nm to $100 \mu\text{m}$

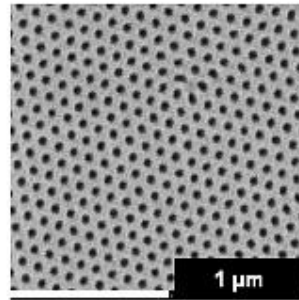
Pore density as high as 10^{11} pores- cm^{-2}

0.5 M H_2SO_4 / 25 V



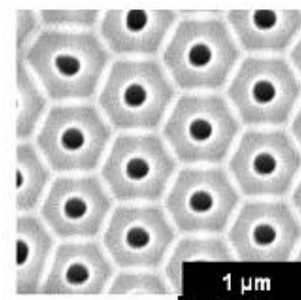
$\varnothing_p \sim 35$ nm, $D_p \sim 60$ nm
 $\sim 3.2 \cdot 10^{10}$ pores/ cm^2

0.5 M $(\text{COOH})_2$ / 40 V



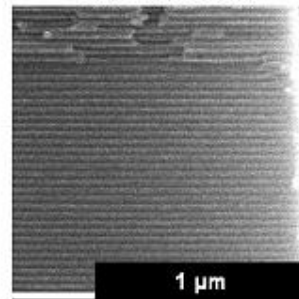
$\varnothing_p \sim 60$ nm, $D_p \sim 100$ nm
 $\sim 1.10^{10}$ pores/ cm^2

1 wt % H_3PO_4 / 195 V



$\varnothing_p \sim 200$ nm, $D_p \sim 450$ nm
 $\sim 6.10^4$ pores/ cm^2

Anodization is a chemical/acid oxidation process, where the metal acts as anode and the acid acts as cathode, providing oxidation species to the anode.



Anodized alumina.
Cagnon, CNRS

Radiation Track-Etch Membranes

Energetic atoms leave
damage track

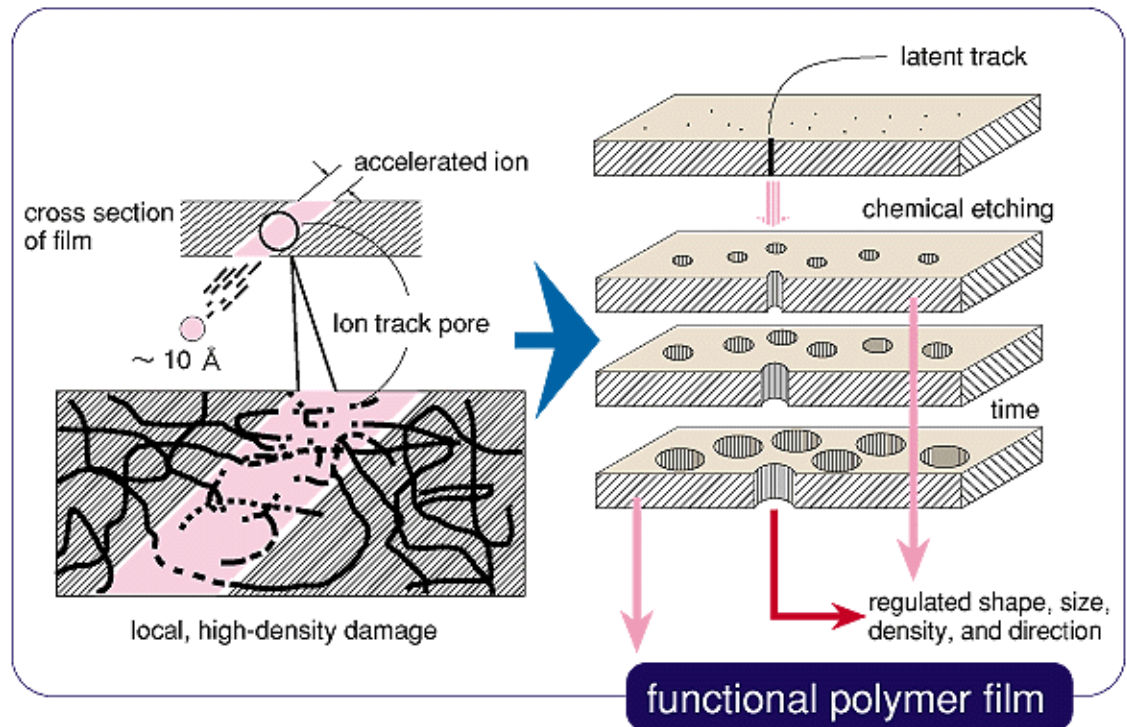
Typical thickness 6-20 μm

Pores with uniform size can
be formed

Pores are randomly
distributed

Pore size as small as 10 nm

Pore density $\sim 10^9$ pores- cm^{-2}



Polymer films with cylindrical pores can be prepared by irradiation of thin polymer (CR-39, 50-100 micrometer) with U (11.6 MeV/n) or Kr (6.2 MeV/n) ions, followed by etching at 60 C in the NaOH solution for 1 hr. Using films of various copolymers with CR-39, pore diameters in the range of 0.5 - 30 micrometer can be obtained.

Japan atomic energy research institute

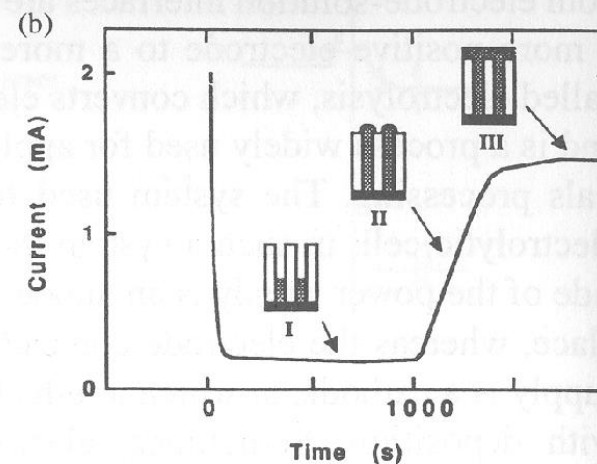
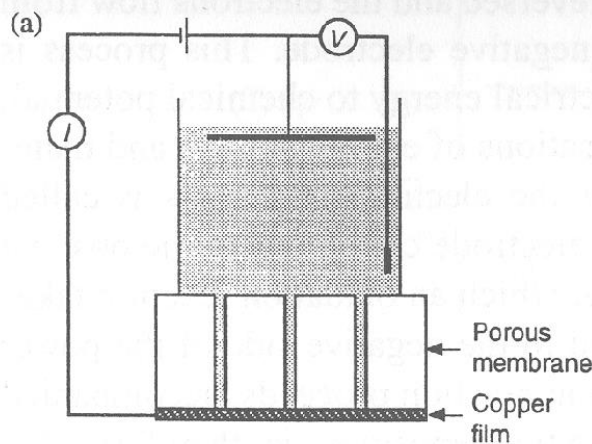
Electrochemical Deposition

Deposition of materials (conductive materials) via electrochemical process, AKA electrodeposition or electroplating:

- Oriented diffusion of charged growth species (cations) through an electric field
- Reduction of charged growth species at the deposition surface (electrode)

Limited to deposition of conductors: metals, alloys, semiconductors and conducting polymers, because deposit must act as electrode after initial deposition

When deposition confined to pores of template membranes → nanocomposites; if template membrane is removed → nanorods and nanowires



Ni deposition into polycarbonate membrane with 60 nm pores at -1V.

Electrophoretic Deposition

Deposition on cathodes from colloidal dispersion

Used for ceramics and organoceramic materials (no need for conducting material)

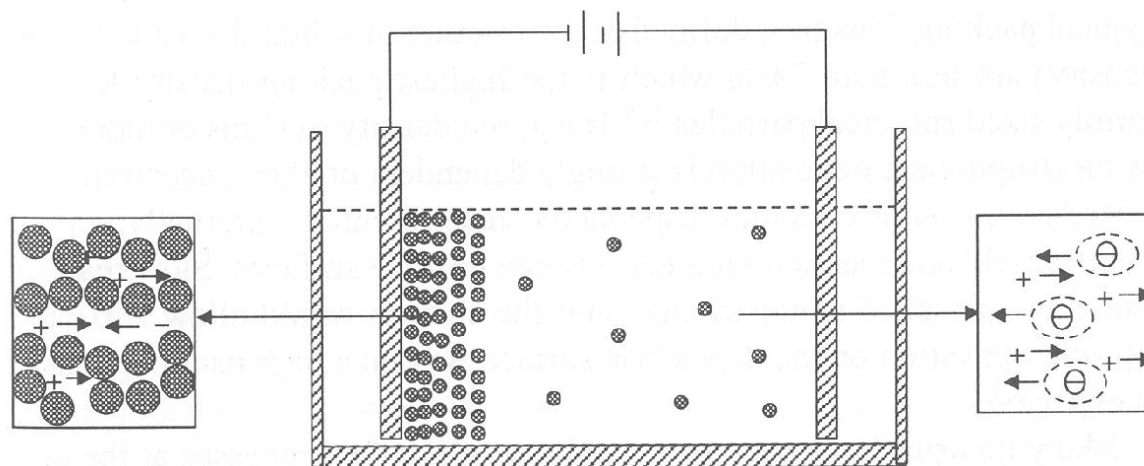
Nanoparticles/nanorods in colloidal dispersion are stabilized (electrostatic/electrosteric)

When placed in a polar solvent or electrolyte solution, electrical charge develops on NP surface

Charged surfaces electrostatically attract oppositely charged species (counter-ions)

When an electric field is applied, the charged particles are set in motion → electrophoresis

Electrophoretic deposition → uses oriented motion of charged particles to grow films on the electrode. If the particles are positive → deposition at cathode



Template Filling

Liquid precursor fills pores directly

Concerns:

- Wettability of pores → complete penetration and filling
- Template material should be chemically inert
- Control of shrinkage during solidification
- Adhesion control → nanorods vs. nanotubules

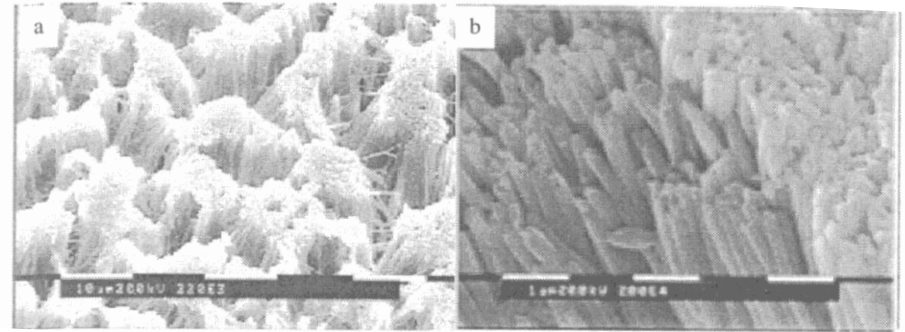
Template Filling

Colloidal dispersion filling

- Soaking in colloidal dispersion
→ filling via capillary forces
- Template removed and heated to high T → removal of template and densification of the colloids (e.g., sintering)

Melt and solution filling

- Filling a template (anodic alumina) with molten metal or polymer solution (injection)
- High pressure is used to assist injection
- E.g., polymers, Se, Te, GaSb, Bi_2Te_3



TiO_2 & ZnO nanorods

- Other methods include:
 - Chemical vapor deposition
 - Deposition by centrifugation

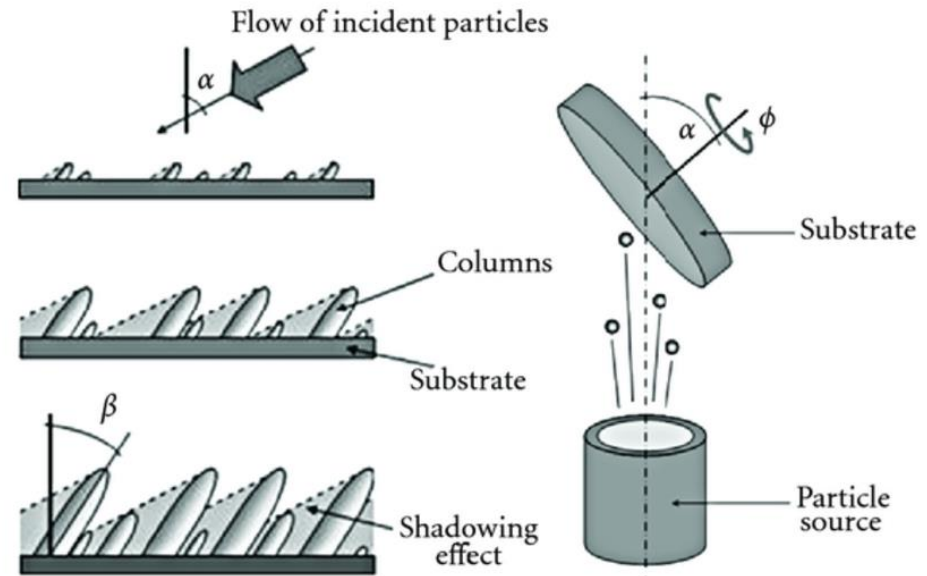
Self-Shadowing Methods

Material deposition from a source in high and ultrahigh vacuum conditions tends to be a “line of sight” process, where atoms travel in a straight line from the source to the substrate

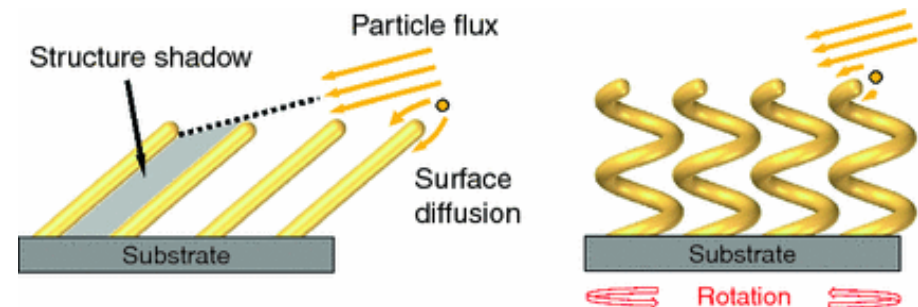
At oblique incidence conditions, a geometric shadow is cast behind topographic features, like islands on a flat surface. This results in a preferential capturing of atoms by topographically prominent features

If surface diffusion is low, the shadowing length dominates the length scale of the features as they grow

Geometric motion of the substrate relative to the source allows for sophisticated control over the geometry of nanoscale features



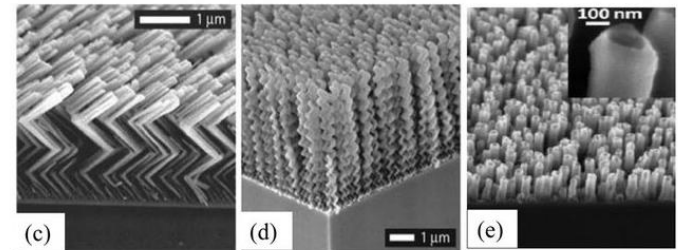
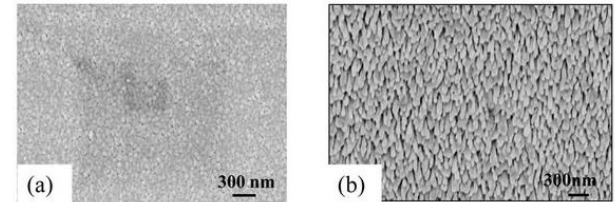
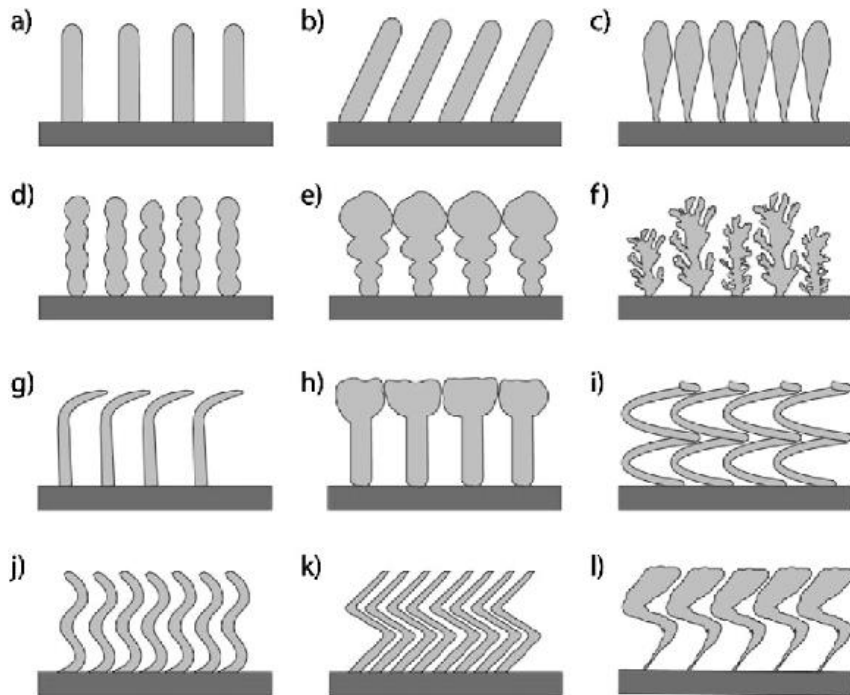
DOI: 10.1155/2012/689154



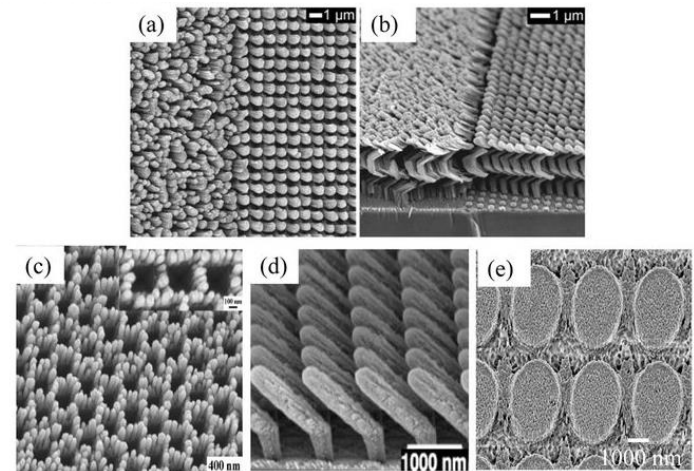
DOI: 10.1007/978-3-642-40128-2

Self-Shadowing Methods

Sample rotation and tilt can be controlled in real time to produce complex shapes for applications ranging from photonics to nanomechanical springs.

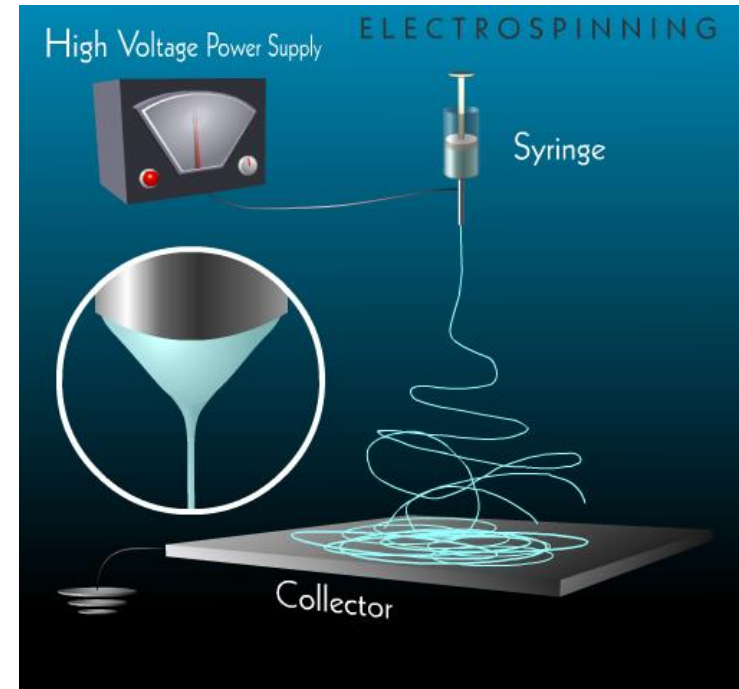


Unpatterned vs pre-patterned surface



Other Means of Making Nanowires

- Chemical conversion of template
 - Nanorods formed by consuming template
- Electrospinning
 - AKA electrostatic fiber processing
 - Principle: electrical force at the surface of polymer overcomes surface tension → electrically charged jet of material is ejected



Michigan Tech

Nanowires via Lithography

